

Number Theory

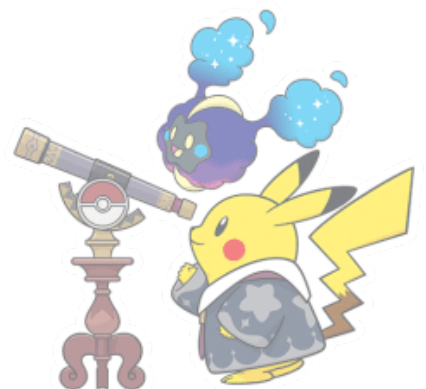
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1 Congruence

1.1 Introduction

Definition. Let $a, b, n \in \mathbb{Z}$ with $n > 0$. We say

$$a \equiv b \pmod{n}$$

if m divides $a - b$, that is,

$$m \mid (a - b)$$

Proposition. Congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Proposition. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

- $a + c \equiv b + d \pmod{m}$
- $ac \equiv bd \pmod{m}$

Proof.

- Since $m \mid (a - b)$ and $m \mid (c - d)$,

$$m \mid ((a - b) + (c - d)) = (a + c) - (b + d)$$

- Since $a = b + km$ and $c = d + \ell m$ for some $k, \ell \in \mathbb{Z}$,

$$ac = bd + m(bl + dk + k\ell m)$$

hence $m \mid (ac - bd)$.

Proposition. If

$$ac \equiv bc \pmod{m}$$

and $\gcd(c, m) = 1$, then

$$a \equiv b \pmod{m}.$$

Proof. From $ac \equiv bc \pmod{m}$ we have $m \mid c(a - b)$. Since $\gcd(c, m) = 1$, it follows that $m \mid (a - b)$, hence $a \equiv b \pmod{m}$.



1.2 Euclidean Algorithm

Definition. Let $a, b \in \mathbb{Z}$ with $b \neq 0$. The **greatest common divisor** of a and b , denoted $\gcd(a, b)$, is the largest positive integer that divides both a and b .

Theorem. For integers a and b with $b \neq 0$ such that

$$a = bq + r, \quad 0 \leq r < |b|$$

Then,

$$\gcd(a, b) = \gcd(b, r)$$

1.3 Congruence Equation

Theorem. (Bézout's Identity) For integers a and b , there exist integers x, y such that

$$ax + by = \gcd(a, b)$$

Proposition. The linear congruence

$$ax \equiv b \pmod{n}$$

has a solution if and only if $\gcd(a, n) \mid b$.

Example. Solve

$$4x \equiv 6 \pmod{10}$$

We have $\gcd(4, 10) = 2$, and $2 \mid 6$, so solutions exist. Then

$$2x \equiv 3 \pmod{5}$$

$$2x \equiv 8 \pmod{5}$$

$$x \equiv 4 \pmod{5}$$



Theorem. (Chinese Remainder Theorem) Let $n_1, n_2, \dots, n_k$ be pairwise coprime positive integers. Then the system

$$\begin{cases} x \equiv a_1 \pmod{n_1} \\ x \equiv a_2 \pmod{n_2} \\ \vdots \\ x \equiv a_k \pmod{n_k} \end{cases}$$

has a unique solution modulo $N = n_1 n_2 \cdots n_k$.

1.4 Fermat's Little Theorem

1.4.1 Introduction

Theorem. (Fermat's Little Theorem) Let p be a prime and let $a \in \mathbb{Z}$. Then

$$a^p \equiv a \pmod{p}$$

If $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof. We first prove that

$$(a+1)^p \equiv a^p + 1 \pmod{p}.$$

By the Binomial Theorem,

$$(a+1)^p = \sum_{k=0}^p \binom{p}{k} a^k = a^p + \binom{p}{1} a^{p-1} + \binom{p}{2} a^{p-2} + \cdots + \binom{p}{p-1} a + 1$$

For each integer k such that $1 \leq k \leq p-1$, the binomial coefficient

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}$$

is divisible by p , since p is prime and does not divide $k!$ or $(p-k)!$. Hence,

$$\binom{p}{k} \equiv 0 \pmod{p} \quad \text{for } 1 \leq k \leq p-1$$



Therefore,

$$(a + 1)^p \equiv a^p + 1 \pmod{p}$$

Now we proceed by induction on a .

Base case: For $a = 0$, we have

$$0^p \equiv 0 \pmod{p},$$

which is true.

Inductive step: Assume that $a^p \equiv a \pmod{p}$ for some integer a . Then,

$$(a + 1)^p \equiv a^p + 1 \equiv a + 1 \pmod{p}.$$

Hence, by induction,

$$a^p \equiv a \pmod{p}$$

holds for all nonnegative integers a .

Example. Compute $3^{100} \pmod{7}$.

Solution. Since 7 is prime and $\gcd(3, 7) = 1$,

$$3^6 \equiv 1 \pmod{7}$$

Write $100 = 6 \cdot 16 + 4$:

$$3^{100} \equiv 3^4 = 81 \equiv 4 \pmod{7}$$

1.4.2 Application

- Fermat's Little Theorem allows large exponents to be reduced modulo $p - 1$:

$$a^n \equiv a^{n \bmod (p-1)} \pmod{p}$$

- (Fermat primality test)** If for some integer a ,

$$a^{p-1} \not\equiv 1 \pmod{p}$$

then p is *not* prime.



2 Quadratic Reciprocity Law

Definition. Let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. We say that a is a **quadratic residue modulo** p if there exists $x \in \mathbb{Z}$ such that

$$x^2 \equiv a \pmod{p}$$

Otherwise, a is called a **quadratic non-residue modulo** p .

Definition. Let p be an odd prime. The **Legendre symbol** $\left(\frac{a}{p}\right)$ is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a quadratic residue mod } p \text{ and } p \nmid a \\ -1, & \text{if } a \text{ is a quadratic non-residue mod } p \\ 0, & \text{if } p \mid a \end{cases}$$

Theorem. (Euler's Criterion) Let p be an odd prime and $p \nmid a$. Then

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$$

Equivalently,

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 1 \pmod{p}, & \text{if } a \text{ is a quadratic residue modulo } p \\ -1 \pmod{p}, & \text{if } a \text{ is a quadratic non-residue modulo } p \end{cases}$$

Theorem. (Supplementary Laws)

- For an odd prime p ,

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

- For an odd prime p ,

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8} \\ -1 & \text{if } p \equiv \pm 3 \pmod{8} \end{cases}$$

Proof. We prove the first supplementary law. By **Euler's criterion**, for any integer a coprime to p ,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$$



Taking $a = -1$, we obtain

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}$$

Now

- If $p \equiv 1 \pmod{4}$, then $\frac{p-1}{2}$ is even, so

$$(-1)^{\frac{p-1}{2}} = 1$$

- If $p \equiv 3 \pmod{4}$, then $\frac{p-1}{2}$ is odd, so

$$(-1)^{\frac{p-1}{2}} = -1$$

Theorem. (Quadratic Reciprocity Law) Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

Example.

- (b) If $p = 2^{4n} + 1$ is an odd prime, show that 7 is a quadratic non-residue of p .
- (c) Determine all primes q such that 5 is a quadratic residue modulo q .
- (d) By considering the natural numbers

$$N = 5(n!)^2 - 1, \quad n = 1, 2, \dots$$

show that there are infinitely many primes of the form $5k - 1$.

- (b) If $p = 2^{4n} + 1$ is an odd prime, show that 7 is a quadratic non-residue of p .
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show that there are infinitely many primes of the form $5k - 1$.



Solution.

(b) Let $p = 2^{4n} + 1$. Then

$$p \equiv 1 \pmod{4}$$

By quadratic reciprocity,

$$\left(\frac{7}{p}\right) = \left(\frac{p}{7}\right)$$

As

$$2^3 \equiv 1 \pmod{7}$$

we have

$$2^{4n} = (2^3)^n \cdot 2^n \equiv 2^n \pmod{7}$$

Checking $2^n \pmod{7}$ shows that

$$2^{4n} + 1 \equiv 3 \text{ or } 5 \pmod{7}$$

However, the quadratic residues modulo 7 are

$$1, 2, 4$$

Hence, 3 and 5 are quadratic non-residues modulo 7. Then

$$\left(\frac{p}{7}\right) = -1$$

and

$$\left(\frac{7}{p}\right) = -1$$

which shows 7 is a quadratic non-residue modulo p .

(c) Let $q \neq 5$ be an odd prime. Since $5 \equiv 1 \pmod{4}$, quadratic reciprocity gives

$$\left(\frac{5}{q}\right) = \left(\frac{q}{5}\right)$$

We want $\left(\frac{q}{5}\right) = 1$. Note that

$$x^2 \equiv 1, 4 \pmod{5}$$



Therefore,

$$\left(\frac{q}{5}\right) = 1 \iff q \equiv 1 \text{ or } 4 \pmod{5}$$

(d) Consider

$$N = 5(n!)^2 - 1$$

Let p be any prime divisor of N . Then

$$5(n!)^2 \equiv 1 \pmod{p}$$

Hence, 5 is a quadratic residue modulo p , and

$$\left(\frac{5}{p}\right) = 1$$

From part (c), this implies

$$p \equiv 1 \text{ or } 4 \pmod{5}$$

However, if $p \equiv 1 \pmod{5}$, fix the prime p and choose $n \geq p$. Then by the definition of the factorial,

$$n! = 1 \cdot 2 \cdot \dots \cdot p \cdot \dots \cdot n$$

so

$$p \mid n!$$

Consequently,

$$p \mid (n!)^2$$

Since $p \mid (n!)^2$, we have

$$(n!)^2 \equiv 0 \pmod{p}$$

Multiplying by 5 yields

$$5(n!)^2 \equiv 0 \pmod{p}$$

Therefore,

$$5(n!)^2 - 1 \equiv -1 \pmod{p}$$

Contradiction.



Hence,

$$p \equiv 4 \pmod{5}$$

i.e.

$$p = 5k - 1$$

Since each N has a prime divisor of this form and N grows without bound, there must be infinitely many primes congruent to $-1 \pmod{5}$.



3 Arithmetic Function

Definition. (Arithmetic Function) An **arithmetic function** is a function

$$f : \mathbb{N} \rightarrow \mathbb{C}$$

Definition. (Euler Totient Function) **Euler's Totient Function**, denoted by $\varphi(n)$, is defined for any positive integer n as

$$\varphi(n) = \#\{k \in \mathbb{Z} \mid 1 \leq k \leq n, \gcd(k, n) = 1\}$$

Lemma. If p is prime, then every integer $1 \leq k \leq p-1$ is coprime to p , hence

$$\varphi(p) = p - 1$$

Lemma. If p is a prime and $k \geq 1$, then

$$\varphi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right)$$

Proof. Among the integers $1 \leq m \leq p^k$, those not coprime to p^k are exactly the multiples of p :

$$p, 2p, 3p, \dots, p^{k-1}p$$

There are p^{k-1} such numbers. Subtracting from the total,

$$\varphi(p^k) = p^k - p^{k-1}$$

Properties. If $\gcd(m, n) = 1$, then

$$\varphi(mn) = \varphi(m)\varphi(n)$$

Theorem. Let

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

be the prime factorization of n . Then

$$\varphi(n) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right)$$



Proof. Using multiplicativity and the prime power formula,

$$\varphi(n) = \prod_{i=1}^r \varphi(p_i^{\alpha_i}) = \prod_{i=1}^r p_i^{\alpha_i} \left(1 - \frac{1}{p_i}\right) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right)$$

Theorem. (Euler's Theorem) If $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

Definition. (Multiplicative Function) An arithmetic function f is called

- **multiplicative** if

$$f(mn) = f(m)f(n) \quad \text{whenever } \gcd(m, n) = 1$$

- **completely multiplicative** if

$$f(mn) = f(m)f(n) \quad \text{for all } m, n$$

Example.

- $\varphi(n)$ is multiplicative but not completely multiplicative.
- $\mu(n)$ is multiplicative.
- $f(n) = n^k$ is completely multiplicative.

Example. Suppose that $f : \mathbb{N} \rightarrow \mathbb{C}$ is multiplicative. Prove that the function

$$g(n) := \sum_{\substack{d|n \\ d \text{ squarefree}}} f(d)$$

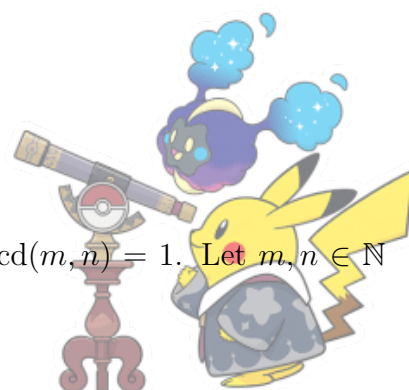
is multiplicative.

Let $k \in \mathbb{Z}$. Compute

$$\sum_{\substack{d|n \\ d \text{ squarefree}}} d^k$$

in terms of the prime factorization $n = \prod_{j=1}^r p_j^{\alpha_j}$.

Solution. Recall that f is multiplicative if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. Let $m, n \in \mathbb{N}$



with $\gcd(m, n) = 1$. Every squarefree divisor d of mn can be written uniquely as

$$d = d_1 d_2$$

where $d_1 \mid m$, $d_2 \mid n$, $\gcd(d_1, d_2) = 1$, and both d_1, d_2 are squarefree. Hence,

$$\begin{aligned} g(mn) &= \sum_{\substack{d \mid mn \\ d \text{ squarefree}}} f(d) \\ &= \sum_{\substack{d_1 \mid m \\ d_1 \text{ squarefree}}} \sum_{\substack{d_2 \mid n \\ d_2 \text{ squarefree}}} f(d_1 d_2) \end{aligned}$$

Since $\gcd(d_1, d_2) = 1$ and f is multiplicative,

$$f(d_1 d_2) = f(d_1) f(d_2)$$

Therefore,

$$\begin{aligned} g(mn) &= \left(\sum_{\substack{d_1 \mid m \\ d_1 \text{ squarefree}}} f(d_1) \right) \left(\sum_{\substack{d_2 \mid n \\ d_2 \text{ squarefree}}} f(d_2) \right) \\ &= g(m) g(n) \end{aligned}$$

Hence, g is multiplicative.

Define $f(d) = d^k$, which is multiplicative. The function

$$g(n) = \sum_{\substack{d \mid n \\ d \text{ squarefree}}} d^k$$

is multiplicative. Hence, it suffices to evaluate $g(p^a)$ for a prime p . The squarefree divisors of p^a are 1 and p . Hence,

$$g(p^a) = 1^k + p^k = 1 + p^k.$$

By multiplicativity,

$$\sum_{\substack{d \mid n \\ d \text{ squarefree}}} d^k = \prod_{j=1}^r (1 + p_j^k).$$

Definition. (Dirichlet Convolution) Given two arithmetic functions f and g , their **Dirichlet convo-**



lution is the arithmetic function

$$(f * g)(n) := \sum_{d|n} f(d) g\left(\frac{n}{d}\right)$$

where the sum runs over all positive divisors d of n .

Example. Let $1(n) = 1$ **constant function**. Then

$$(1 * 1)(n) = \sum_{d|n} 1 = \tau(n)$$

which is the number of divisors of n .

Definition. (Möbius Function) The **Möbius function** $\mu(n)$ is defined as

$$\mu(n) = \begin{cases} 1, & n = 1 \\ (-1)^k, & n \text{ is a product of } k \text{ distinct primes} \\ 0, & n \text{ has a squared prime factor} \end{cases}$$

Lemma. For all $n \in \mathbb{N}$,

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1 \\ 0, & n > 1 \end{cases}$$

Proof. If $n = 1$, the only divisor is 1, so the sum equals $\mu(1) = 1$. Now let $n > 1$ and write its prime factorization as

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

Only square-free divisors contribute to the sum, so

$$\sum_{d|n} \mu(d) = \sum_{S \subseteq \{1, 2, \dots, r\}} (-1)^{|S|} = (1 - 1)^r = 0$$

Hence, by definition of Dirichlet convolution,

$$(\mu * 1)(n) = \sum_{d|n} \mu(d) 1\left(\frac{n}{d}\right)$$

Since $1(k) \equiv 1$ for all k , this simplifies to

$$(\mu * 1)(n) = \sum_{d|n} \mu(d)$$



From above,

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1 \\ 0, & n > 1 \end{cases}$$

Therefore,

$$(\mu * 1)(n) = \varepsilon(n)$$

Example. Suppose that $f : \mathbb{N} \rightarrow \mathbb{R}$ is an arithmetic function such that

$$\sum_{n \in \mathbb{N}} |f(n)|$$

converges. Let $N \in \mathbb{N}$. Prove that

$$\sum_{\substack{n \in \mathbb{N} \\ \gcd(n, N) = 1}} f(n) = \sum_{d|N} \mu(d) \sum_{n \in \mathbb{N}} f(dn).$$

Solution. From the above identity,

$$\begin{aligned} \sum_{\substack{n \in \mathbb{N} \\ \gcd(n, N) = 1}} f(n) &= \sum_{n \in \mathbb{N}} f(n) \sum_{d|\gcd(n, N)} \mu(d) \\ &= \sum_{d|N} \mu(d) \sum_{\substack{n \in \mathbb{N} \\ d|n}} f(n). \end{aligned}$$

Note that as $\sum |f(n)|$ converges, we may interchange sums. Writing $n = dm$,

$$\sum_{\substack{n \in \mathbb{N} \\ d|n}} f(n) = \sum_{m \in \mathbb{N}} f(dm)$$

Therefore,

$$\sum_{\substack{n \in \mathbb{N} \\ \gcd(n, N) = 1}} f(n) = \sum_{d|N} \mu(d) \sum_{m \in \mathbb{N}} f(dm)$$

Definition. (Identity Function) Define the identity function ε by

$$\varepsilon(n) = \begin{cases} 1, & n = 1 \\ 0, & n > 1 \end{cases}$$



Theorem. (Möbius Inversion Formula) Let f and g be arithmetic functions. If

$$g(n) = \sum_{d|n} f(d)$$

then

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right)$$

Proof. The relation

$$g(n) = \sum_{d|n} f(d)$$

can be written using Dirichlet convolution as

$$g = f * 1$$

Convolving both sides with μ , we obtain

$$\mu * g = \mu * (g * 1)$$

Since Dirichlet convolution is associative, this becomes

$$\mu * g = (\mu * 1) * f$$

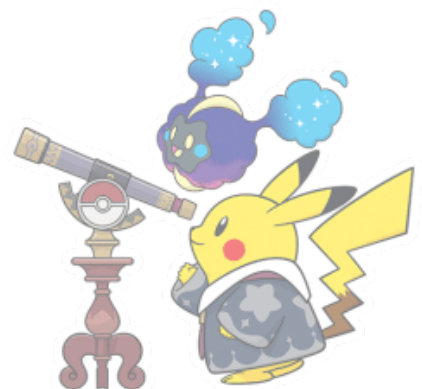
By the lemma, $\mu * 1 = \varepsilon$, hence

$$\mu * g = \varepsilon * f = f$$

Expanding the convolution gives

$$f(n) = (\mu * g)(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right)$$

which completes the proof.



4 Multiplicative Order

Definition. Let $n \geq 2$ and let a be an integer with $\gcd(a, n) = 1$. The **order of a modulo n** , denoted $\text{ord}_n(a)$, is the smallest positive integer k such that

$$a^k \equiv 1 \pmod{n}$$

Properties. Let $d = \text{ord}_n(a)$. Then

If

$$a^k \equiv 1 \pmod{n},$$

then

$$d \mid k$$

Proof. Write $k = qd + r$ with $0 \leq r < d$. Then

$$a^k \equiv (a^d)^q a^r \equiv a^r \pmod{n}$$

If $a^k \equiv 1 \pmod{n}$, minimality of d forces $r = 0$.

Example. Let p be an odd prime. Suppose $n \in \mathbb{N}$ satisfies

$$\gcd(n, p^2 - 1) = 1 \quad \text{and} \quad n \mid (p^n + 1)$$

Prove that $n = 1$.

From $n \mid (p^n + 1)$, we obtain

$$p^n \equiv -1 \pmod{n}$$

Squaring gives

$$p^{2n} \equiv 1 \pmod{n}$$

Hence, the order of p modulo n divides $2n$ but not n , so it must divide 2. Therefore,

$$p^2 \equiv 1 \pmod{n}$$

Therefore, $n \mid (p^2 - 1)$. Since $\gcd(n, p^2 - 1) = 1$, we conclude $n = 1$.



5 Introduction to Continued Fraction

Definition. A **simple continued fraction** (SCF) is an expression of the form

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \ddots}}}$$

where

$$a_0 \in \mathbb{Z}, \quad a_1, a_2, a_3, \dots \in \mathbb{N}$$

We use the compact notation

$$[a_0; a_1, a_2, a_3, \dots]$$

Theorem.

- If the sequence terminates, i.e. $[a_0; a_1, \dots, a_n]$, the value is a **rational number**.
- If the sequence is infinite, the value is an **irrational number**.

Proof. We prove the first part only. Let

$$x = \frac{p}{q}, \quad p, q \in \mathbb{Z}, \quad q > 0.$$

Apply the Euclidean Algorithm:

$$p = a_0q + r_1$$

$$q = a_1r_1 + r_2$$

$$r_1 = a_2r_2 + r_3$$

$$\vdots$$

$$r_{n-2} = a_{n-1}r_{n-1} + r_n$$

$$r_{n-1} = a_nr_n$$

Then

$$\frac{p}{q} = [a_0; a_1, a_2, \dots, a_n]$$



Example. Find the continued fraction of $\frac{43}{19}$.

Solution.

$$43 = 2 \cdot 19 + 5$$

$$19 = 3 \cdot 5 + 4$$

$$5 = 1 \cdot 4 + 1$$

$$4 = 4 \cdot 1$$

Hence,

$$\frac{43}{19} = [2; 3, 1, 4]$$

Example.

(a) Consider the repeated infinite regular continued fraction

$$\alpha = [\overline{a, b}], \quad a, b \in \mathbb{N}$$

Writing

$$\alpha = A + B\sqrt{D}$$

with $A, B \in \mathbb{Q}$ and $D \in \mathbb{Z}$ squarefree, prove that D depends only on the product ab and not on the individual values of a and b .

(b) Consider the continued fraction $\alpha = [\overline{3, 7}]$, written as in part (a). Find all odd primes p for which there exists an element of $\mathbb{Z}/p\mathbb{Z}$ satisfying the same quadratic equation as α . In other words, suppose that

$$A\alpha^2 + B\alpha + C = 0, \quad A, B, C \in \mathbb{Z}, \quad \gcd(A, B, C) = 1$$

Find all odd primes p for which

$$Ax^2 + Bx + C \equiv 0 \pmod{p}$$

is solvable.



Solution. Let

$$\alpha = [\overline{a, b}] = a + \frac{1}{b + \frac{1}{a + \frac{1}{b + \ddots}}}$$

By periodicity, the tail after the first two entries is again α , so we obtain

$$\alpha = a + \frac{1}{b + \frac{1}{\alpha}}$$

Rewriting,

$$\alpha - a = \frac{1}{b + \frac{1}{\alpha}} \implies \frac{1}{\alpha - a} = b + \frac{1}{\alpha}$$

Hence

$$\frac{1}{\alpha - a} - \frac{1}{\alpha} = b$$

Combining the fractions gives

$$\frac{a}{\alpha(\alpha - a)} = b$$

or equivalently

$$ab = \frac{a^2b}{a} = \frac{a}{\alpha(\alpha - a)} \cdot a \implies \alpha^2 - a\alpha - \frac{a}{b} = 0$$

Multiplying through by b yields the quadratic equation

$$b\alpha^2 - ab\alpha - a = 0$$

The discriminant of this quadratic is

$$\Delta = (ab)^2 + 4ab = ab(ab + 4)$$

Hence

$$\alpha = \frac{ab + \sqrt{ab(ab + 4)}}{2b} = A + B\sqrt{D}$$

where D is the squarefree part of $ab(ab + 4)$.

For $\alpha = [\overline{3, 7}]$, we have $ab = 21$. From part (a), α satisfies

$$7\alpha^2 - 21\alpha - 3 = 0$$



Thus $A = 7$, $B = -21$, $C = -3$, and the discriminant is

$$\Delta = (-21)^2 - 4(7)(-3) = 441 + 84 = 525 = 21 \cdot 25.$$

The squarefree part is therefore

$$D = 21 = 3 \cdot 7.$$

For an odd prime p , the congruence

$$7x^2 - 21x - 3 \equiv 0 \pmod{p}$$

is solvable if and only if the discriminant Δ is a quadratic residue modulo p , i.e.

$$\left(\frac{\Delta}{p}\right) = 1$$

Since $\Delta = 25 \cdot 21$ and 25 is always a quadratic residue modulo odd p , this condition reduces to

$$\left(\frac{21}{p}\right) = 1$$

For $p \nmid 21$, we factor

$$\left(\frac{21}{p}\right) = \left(\frac{3}{p}\right) \left(\frac{7}{p}\right)$$

Using quadratic reciprocity,

$$\left(\frac{3}{p}\right) = 1 \iff p \equiv \pm 1 \pmod{12},$$

and

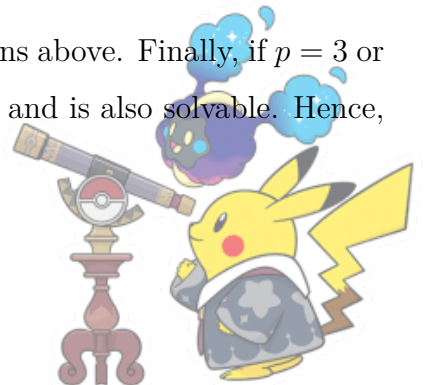
$$\left(\frac{7}{p}\right) = 1 \iff p \equiv 1, 2, 4 \pmod{7}.$$

Therefore, for odd primes $p \neq 3, 7$, the congruence is solvable if and only if

$$\left(\frac{21}{p}\right) = 1,$$

that is, if p lies in one of the residue classes modulo 84 satisfying both conditions above. Finally, if $p = 3$ or $p = 7$, then $\Delta \equiv 0 \pmod{p}$, so the congruence has a repeated root modulo p and is also solvable. Hence, the odd primes p for which

$$7x^2 - 21x - 3 \equiv 0 \pmod{p}$$



is solvable are exactly those odd primes p such that

$$\left(\frac{21}{p}\right) = 1$$

including $p = 3$ and $p = 7$.

